Between the Lines: a computational analysis of migration narrative in The New York Times

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Course in Computational Social Science

GitHub link: https://github.com/lisaprem/CSS-NYT-migration-analysis

Abstract

This study presents a computational analysis of the migration narrative in *The New York Times* from 2005 to 2025. Using a dataset of articles retrieved via the NYT Archive API, the research applies a mixed-methods approach combining exploratory lexical analysis, sentiment analysis (BING and AFINN lexicons), Latent Dirichlet Allocation (LDA) for topic modelling and statistical regression models. The findings indicate that the coverage of migration exhibits consistently a negative sentiment. Quantitative results identify the specific thematic focus, particularly border security and law enforcement, as the primary predictor of the article's emotional tone. Furthermore, the empirical models demonstrate a negative correlation between article length and sentiment: increased word count correlates systematically with lower sentiment scores. Finally, the data shows longitudinal editorial stability. The core sentiment exhibits no statistically significant shifts across the Bush, Obama, Trump and Biden administrations, indicating that the institutional framing of migration remains stable, independent of political cycles.

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1.Introduction

1.1 The impact of media language

In contemporary societies, mass media play a central role in shaping public understanding of complex social phenomena. Through the selection of topics, the framing of issues and the language used to describe events, media outlets contribute to constructing narratives that influence how audiences interpret political and social developments and which topics to focus on. As highlighted by the theory of agenda-setting and framing, media institutions do not simply report events but also influence how issues are perceived by emphasizing particular aspects of reality while downplaying others (McCombs, 2004; Entman, 1993). Even when information is effectively accurate, the tone and framing of reporting can affect readers' interpretations and emotional responses.

For this reason, scholars in media and communication studies have increasingly examined the role of language and narrative structures in shaping public opinion. Traditional approaches to studying media discourse have often relied on qualitative content analysis or manual coding of texts, which allow for detailed interpretation of framing strategies but are difficult to apply to large textual corpora or long time spans (Neuendorf, 2017). The growing availability of digital news archives, however, has opened new possibilities for analysing media discourse using computational approaches. As a result, these approaches have become an increasingly valuable tool for monitoring discourse within public information systems and assessing how language is used to frame sensitive issues.

1.2 Migration as a growing and multi-dimensional issue

The study of media narratives becomes particularly relevant when examining topics characterized by strong political and social implications. Among these, migration represents one of the most debated and contested issues in contemporary public discourse.

Job opportunities, conflicts or natural disasters, people have been constantly on the move for ages, whether voluntarily or due to circumstances. The twenty- first century has been called "the age of migration", essentially because there are more migrants in the world today than ever before: about 232 million international migrants in 2013 (UK Parliament, 2024). In general, it is expected that the number of international migrants worldwide will increase in the future, with demographic factors, economic disparities and environmental change continuing to be major drivers of migration.

Migration intersects with multiple dimensions of social life, including humanitarian concerns, labour market dynamics, border security, demographic change and national identity (Castles, de Haas, Miller, 2014). Because of this multidimensional nature, migration is often interpreted through different and sometimes competing perspectives, ranging from humanitarian narratives to security-oriented or economic frames.

Previous research has shown that media representations of migration frequently rely on recurring narrative frames that shape public perceptions of the phenomenon. Studies of migration discourse have identified common frames such as crisis, security threat, humanitarian emergency or economic opportunity (Benson, 2013; Berry, Garcia-Blanco, Moore, 2015). These representations can influence public attitudes toward immigrants and

migration policies by emphasizing particular aspects of the issue while marginalizing others. As a consequence, understanding how migration is portrayed in the media is essential for analysing how public narratives surrounding migration are constructed.

1.3 Migration in the United States

The United States provides a particularly relevant context in which to investigate these dynamics. Historically, migration has played a fundamental role in shaping American society, contributing to its demographic diversity and economic development. At the same time, migration has long been a subject of political debate and social contestation. In recent decades, the issue has become increasingly important in political discourse, at the centre of electoral campaigns, policy debates and media coverage (Massey et al., 2015). The politicization of immigration has also contributed to strong polarization in public attitudes, making the topic particularly sensitive within media narratives.

In addition to its political relevance, the United States offers a good background for longitudinal analysis due to the extensive availability of digital news archives. Major newspapers maintain large databases of journalistic content that span several decades, allowing researchers to investigate how media narratives evolve across different political administrations and historical contexts. Such longitudinal analyses make it possible to assess whether media coverage remains relatively stable over time or whether it shifts in response to political changes or other events.

1.4 Research questions

Given the central role of media communication in growing the interest and shaping public understanding of complex social issues, and considering that migration represents one of the most debated and politically sensitive topics in contemporary societies, particularly in the United States, this study aims to address three main research questions.

First, the analysis investigates how a major media institution such as *The New York Times* approaches a sensitive topic like migration. By examining the language and thematic structures present in migration-related articles, the study seeks to identify the dominant narrative frames and the overall tone that characterize the newspaper's coverage of this issue.

Second, the study explores how this narrative has evolved over the past two decades, focusing on the period between 2005 and 2025. Media narratives are not static but tend to change in response to political developments, events with a strong impact and shifts in public debate. Analysing the temporal evolution of the discourse therefore makes it possible to assess whether the framing and emotional tone of migration coverage have remained relatively stable or have undergone significant transformations over time.

Finally, the study examines whether the narrative promoted by *The New York Times* has been influenced by broader political contexts, particularly by changes in presidential administrations. Given the strong politicization of migration in the United States, it is relevant to investigate whether different political environments correspond to variations in the sentiment and framing of migration-related reporting.

By addressing these questions through computational text analysis, this research aims to provide a data-driven perspective on how migration has been portrayed in a major American newspaper over the past twenty years.

2. Data

2.1 Data collection

The data used in this study were collected from the New York Times API, an official interface that provides structured access to metadata and article summaries published by the newspaper. The NYT offers several APIs to access its content: the Archive API, which allows access to all articles published in a given month and year; the Article Search API for keyword, author, date or section queries; the Books API for bestseller and review information; the Most Popular API for the most read or shared articles; the Times Wire API and RSS Feeds for real-time updates; and the Top Stories API for currently featured articles in each section. I specifically retrieved the data through the Archive API, which is particularly useful for large-scale historical analyses.

The analysis focuses on articles published between 2005 and 2025, a twenty-year period chosen to capture long-term developments in media discourse on migration. This time span allows the investigation of narrative continuity and change across multiple political cycles and presidential administrations, providing sufficient temporal depth for longitudinal analysis.

From each archived article, a set of core variables was extracted: the selected information includes the publication date, the headline, the abstract, the author and the news section. The publication date enables the examination of trends over time, the news section provides contextual information about where migration-related topics are positioned within the newspaper's editorial structure, the headline captures the immediate framing of the news story and the abstract serves as the primary textual unit of analysis.

To identify articles specifically related to migration, a keyword-based filtering procedure was applied. The filtering relied on a set of migration-related terms, including *"migration", "migrations", "immigrant", "immigrants", "migrate"* and *"immigrate"*. To increase both coverage and selection reliability, the filter was applied to both the article headline and the abstract. This dual approach reduces the risk of excluding relevant articles that discuss migration extensively but mention it only in one textual field.

2.2 Data cleaning

After data retrieval, a structured data cleaning process was implemented to ensure consistency across the corpus and to minimize noise introduced by archival formatting.

First, entries with missing abstracts were removed. During a preliminary inspection, it emerged also that in earlier years of the archive some abstract fields contained multiple article summaries concatenated within a single entry. These merged abstracts were systematically separated by sequences of three or more blank spaces. To ensure that content referring to unrelated topics was not included in the dataset, abstracts were split where these blank spaces

occurred, allowing the transformation of aggregated text fields into single abstracts. The dataset was then filtered again based on migration-related keywords.

Next, several noise-removal procedures were applied. Archival artifacts and metadata automatically included in the NYT archive, such as page indicators (e.g., *PAGE A1*) or section labels, were filtered out to prevent non-editorial content from influencing lexical and sentiment analyses. Duplicate or structurally irrelevant entries were also excluded when identified.

To reduce potential bias associated with document length, an outlier removal step was introduced. Abstracts exceeding 200 words were discarded, as these entries typically corresponded to transcripts, aggregated daily summaries or non-standard article formats. Controlling for excessive length is particularly important in sentiment analysis, where longer texts may amplify polarity scores.

2.3 Data preprocessing

Following data cleaning, an additional preprocessing stage was conducted to prepare the textual corpus for sentiment analysis and topic modelling.

Each article abstract was first assigned a unique identifier to preserve the structure throughout the transformation process. The text was then tokenized into individual words using *unnest_tokens*() from tidytext**.**

The resulting list of words was further cleaned through a three-step filtering procedure. First, single-character tokens and numeric values were removed, as they do not convey meaningful semantic information. Second, non-informative words were excluded using two complementary stop word lists. The first comprised standard English stop words from the *stopwords* package, which filter out common function words (e.g. “the”, “and”, “is”) with minimal semantic value. The second was a custom, context-specific list designed to remove terms frequently appearing in news articles but irrelevant to migration-related content. This list included general reporting terms (e.g., “said”, “reported”, “wrote”), temporal references (e.g., “today”, “yesterday”, “month”), *NYT*-specific terms (e.g., “crossword”, “newsletter”, “podcast”), general verbs (e.g., “make”, “take”, “see”) and numeric/quantity words (e.g., “million”, “hundred”, “four”).

This multi-step filtering ensured that the remaining vocabulary primarily reflected thematic content rather than stylistic or institutional language patterns. The resulting tokenized corpus served as input for the following sentiment analysis and Latent Dirichlet Allocation (LDA) topic modeling.

3.Methodology

3.1 Exploratory lexical analysis

As an initial step, an exploratory lexical analysis was conducted to obtain an overview of the vocabulary and thematic structures within the corpus. A word cloud was used to visualize the most frequent terms appearing across article abstracts, providing an intuitive representation of

salient topics and recurring frames in migration coverage. In addition, a bigram network analysis was performed to examine co-occurring word pairs, allowing the identification of commonly linked concepts and policy-related expressions. Together, these exploratory techniques offer a descriptive assessment of linguistic patterns in the dataset and help contextualize subsequent sentiment analysis and topic modelling results.

3.2 Sentiment analysis

Sentiment analysis is a computational technique used to identify and quantify the emotional tone in textual data. In the context of media and newspaper analysis, sentiment analysis represents a valuable tool because it allows researchers to examine not only what news articles report, but also how they communicate it. News coverage conveys meaning through language choices, tone and emotional emphasis, which can influence how topics are perceived by audiences. By analysing sentiment, researchers can explore how issues are framed, whether coverage tends to highlight concern, conflict or optimism, and how these patterns change across large collections of articles over time.

Following preprocessing and tokenization, each tokenized word was matched with specific sentiment dictionaries. The overall sentiment of an abstract is then estimated by aggregating these word-level scores. The scores were assigned using two different dictionaries in order to capture both the direction and the intensity of emotional expression and to increase the robustness of the analysis:

- The first dictionary employed was the Bing Liu lexicon, originally developed within early research on opinion mining by Bing Liu. This lexicon categorizes words as either positive or negative, allowing overall sentiment to be estimated by comparing the relative frequency of positive and negative terms within each abstract. Its binary structure provides a transparent measure of general evaluative tone and has been widely applied in studies analysing public discourse and media narratives.
- The second resource, the AFINN lexicon, created by Finn Årup Nielsen, assigns numerical sentiment scores ranging from −5 to +5 to individual words. This approach captures not only the direction but also the intensity of emotional expression. By aggregating word-level scores across texts, the method produces a continuous sentiment measure that reflects variations in emotional strength across documents.

Because abstracts vary substantially in length, raw sentiment scores may partly reflect differences in word counts rather than substantive variation in tone. Sentiment scores were therefore normalized by text length to ensure comparability across abstracts and to better capture relative sentiment intensity. Specifically, for each abstract *i*, the normalized sentiment score was calculated as:

$$S_i^{norm} = \frac{\sum_{j=1}^{n_i} s_{ij}}{n_i}$$

Formula 1: normalized sentiment score

where s_{ij} represents the sentiment value assigned to word j in abstract i, and n_i denotes the total number of tokens in that abstract.

3.3 Topic modelling (LDA)

Latent Dirichlet Allocation (LDA) is a probabilistic model for topic modelling. It assumes that each document can be represented as a mixture of latent topics and that each topic is a distribution over words. By estimating these two distributions simultaneously, LDA uncovers the underlying thematic structure within a corpus of texts.

In this study, LDA was applied to the preprocessed abstracts. The trimmed Document–Term Matrix, generated from tokenized and cleaned text, was used as input to the model. LDA was estimated using Gibbs sampling with a predefined number of five topics and 2000 iterations, with a fixed random seed for reproducibility. Posterior topic probabilities were computed for each document and the dominant topic was determined as the topic with the highest probability.

3.4 Statistical modelling

3.4.1 Multiple Linear Regression model

To estimate the extent to which each explanatory variable affects the sentiment score, a multiple linear regression framework is employed. This approach allows the effect of each predictor to be evaluated while holding the other variables constant.

Two model specifications are estimated. In the first model, the sentiment score is a function of the following predictors: word count**,** year and presidential administration.

$$sentiment_mean_i = \beta_0 + \beta_1 year_i + \beta_2 president_i + \beta_3 word_count_i + \epsilon_i$$

Formula 2: multiple linear regression model, 1

In a second model, the dominant topic identified through topic modelling is additionally included as an explanatory variable.

$$sentiment_mean_i = \beta_0 + \beta_1 year_i + \beta_2 president_i + \beta_3 word_count_i + \beta_4 topic_i + \epsilon_i$$

Formula 3: multiple linear regression model, 2

By evaluating both models, this approach allowed us to assess how much explanatory power (R^2) is gained when shifting from purely external factors (time and politics) to the specific subject matter being covered.

3.4.2 Test ANOVA

To further evaluate the relative importance of the predictors, an Analysis of Variance (ANOVA) test is conducted on the regression models. The null hypotheses tested are:

$$H_0 : \mu_{Bush} = \mu_{Obama} = \mu_{Trump} = \mu_{Biden}$$
$$H_0 : \beta_{year} = 0$$
$$H_0 : \beta_{word_count} = 0$$
$$H_0 : \mu_{Topic1} = \mu_{Topic2} = \mu_{Topic3} = \mu_{Topic4} = \mu_{Topic5}$$

ANOVA decomposes the total variance of the dependent variable into the portions attributable to each explanatory factor, allowing the assessment of whether the inclusion of specific variables significantly improves the explanatory power of the model.

$$SS_{Total} = SS_{Model} + SS_{Residuals}$$
$$F = \frac{MS_{Model}}{MS_{Residuals}} = \frac{SS_{Model}/df_{Model}}{SS_{Residuals}/df_{Residuals}}$$

Formula 4: test ANOVA

In this study, the ANOVA test is used to compare the contribution of variables to the overall variation in sentiment scores. While the regression coefficients indicate the direction and magnitude of the relationships, the ANOVA results help determine which factors account for the largest share of variation in the sentiment of the coverage.

4. Results

4.1 Exploratory analysis

The exploratory lexical analysis provides an initial overview of how migration is discussed in the *New York Times* across the period 2005–2025. By examining the most frequent words and word combinations, it is possible to identify the dominant themes shaping the narrative.

The distribution of the most frequent words reveals a strong emphasis on legality and border-related issues. The terms *illegal* and *border* appear among the most frequent in the corpus, indicating that migration is often framed through the lens of irregular migration and territorial control rather than purely humanitarian or demographic perspectives.

Figure 1: Word cloud. (Made by https://www.wordclouds.com/)

In addition to this legal framing, the lexical distribution also reflects the political dimension of the migration debate. Words such as *policy*, *law*, *federal* and references to political actors

appear frequently in the dataset. This suggests that migration coverage is closely linked to the political discourse.

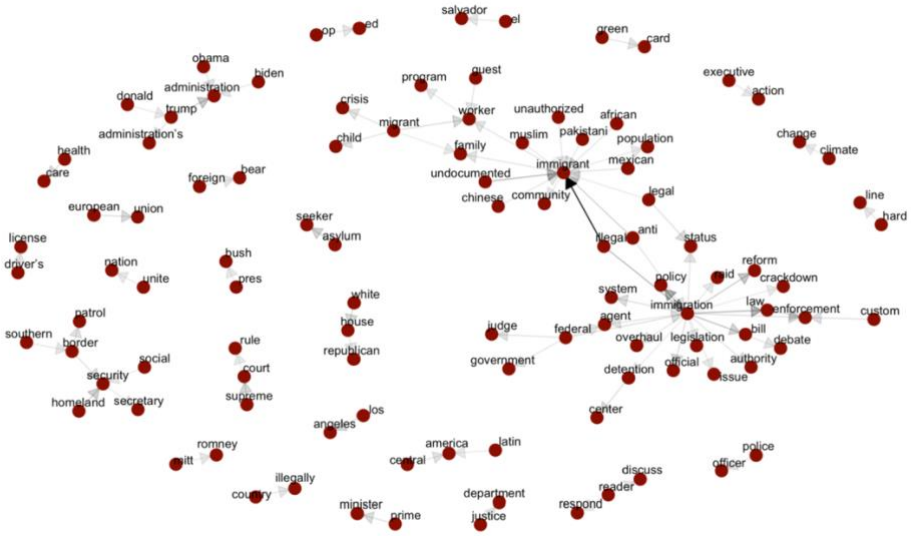


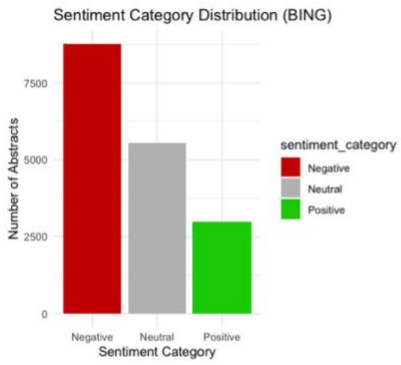
Figure 2: Bigrams network.

The analysis of bigrams further clarifies these patterns. The most common word combination is *illegal immigrant*, which strongly reinforces the centrality of legal status in the newspaper's narrative. Other frequent bigrams include expressions related to policy and governance, such as *immigration policy*, *immigration reform* and *immigration law*. These associations confirm that migration is frequently discussed in terms of legislation and regulatory frameworks. Some bigrams also point to the presence of humanitarian and security-related issues, such as *asylum seeker* and *homeland security*. These expressions reflect the broader geopolitical context in which migration is debated, particularly in relation to security concerns and international crises.

Overall, the exploratory analysis suggests that the *New York Times* migration narrative is structured around three main dimensions: border control, legal status, and political debate**.** Human-centered perspectives appear in the dataset but play a less prominent role in the dominant vocabulary.

4.2 Sentiment analysis results

The sentiment analysis reveals a clear imbalance between negative and positive language in the coverage of migration.



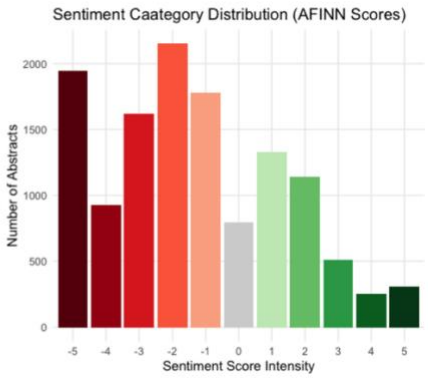


Figure 3: Sentiment category distribution (BING)

Figure 4: Sentiment category distribution (AFINN)

As shown in Figure 3, the distribution of sentiment categories based on the Bing lexicon reveals a clear predominance of negative abstracts. Negative texts represent the largest share of the dataset, followed by neutral abstracts, while positive texts account for the smallest proportion. This pattern suggests that migration is more frequently discussed using language associated with problems, conflicts, or institutional challenges rather than positive developments.

A similar pattern emerges from the AFINN lexicon. Figure 4 presents the distribution of sentiment intensity scores, which range from −5 to +5. The histogram shows a higher concentration of words with negative scores, particularly around −2 and −3, while positive scores are less frequent and more evenly distributed across lower intensity levels. This distribution indicates that negative sentiment is not only more common but also tends to be expressed with stronger lexical intensity.

Further insights emerge from the analysis of the most influential words contributing to sentiment.

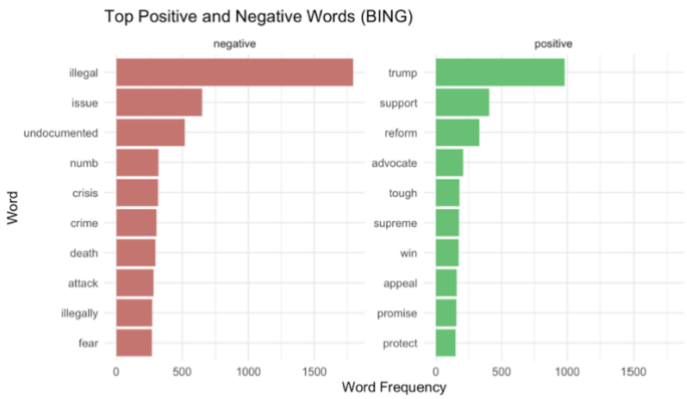


Figure 5: Top positive/negative words (BING)

Figure 5 displays the most frequent positive and negative words identified by the Bing lexicon. Among negative terms, the word *illegal* stands out as the most frequent, followed by terms such as *undocumented*, *crisis*, and *crime*. These words suggest that migration is often framed through themes of legality, security and social problems. In contrast, positive words such as *support*, *reform* and *protect* appear less frequently and are often associated with policy initiatives or advocacy efforts rather than with migrants themselves.

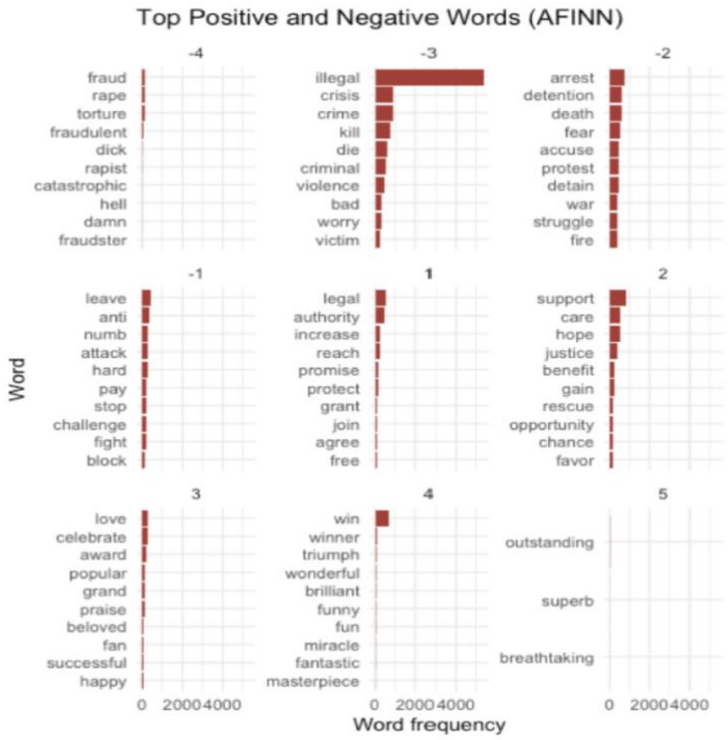


Figure 6: Top positive/negative words (AFINN)

The results based on the AFINN lexicon, shown in Figure 6**,** provide additional information about the intensity of sentiment**.** Words with strong negative scores include *fraud*, *catastrophic* and *disaster*, which contribute to highly negative sentiment values. Words with moderately negative scores, such as *crisis*, *crime* and *illegal*, appear more frequently, indicating that much of the negative tone is driven by recurring policy and security terminology. On the positive side, words such as *support*, *protect* and *opportunity* appear with moderate positive scores, while strongly positive words (for example *outstanding* or *superb*) are rare in the dataset. Overall, these patterns confirm that negative sentiment is both more frequent and more lexically dominant in migration-related abstracts.

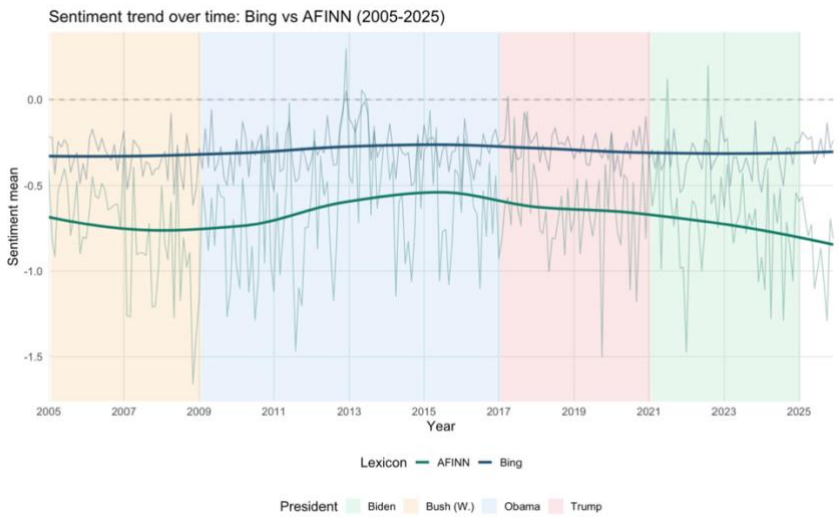


Figure 7: Sentiment trend over time: BING vs AFINN

The temporal evolution of sentiment is illustrated in Figure 7, which compares the average sentiment over time using both the Bing and AFINN lexicons. Despite short-term fluctuations, the overall trend remains relatively stable throughout the period from 2005 to 2025. Both measures consistently indicate a slightly negative average sentiment.

The figure also highlights the different presidential administrations during the study period. However, no clear shifts in sentiment appear to coincide with these political transitions. Although minor variations occur across years, the general tone of migration coverage remains broadly consistent over time. This suggests that the overall framing of migration in the newspaper is relatively stable and not strongly driven by changes in political leadership.

To further examine the determinants of sentiment, a multiple linear regression model was estimated (Figure 8).

Predictor	Estimate (β)	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	1.6713	7.9155	0.211	0.833	
Year	-0.0009	0.0039	-0.246	0.806	
President (Obama)	-0.0176	0.0305	-0.578	0.563	
President (Trump)	0.0256	0.0511	0.502	0.616	
President (Biden)	-0.0772	0.0685	-1.126	0.260	
President (Trump II)	0.0103	0.0784	0.132	0.895	
Word Count	**-0.0307**	**0.0033**	**-9.129**	**< 2e-16**	*******

F-statistic: $p < 2.2*10^{(-16)}$ - **Adjusted R^2**: 0.008

Figure 8: Multiple linear regression model summary

The results show that word count is the only statistically significant predictor of sentiment. The negative coefficient indicates that longer abstracts are associated with more negative sentiment. This suggests that more detailed articles tend to focus on complex or problematic aspects of migration.The presidential variables are not statistically significant compared to the Bush administration used as the baseline. This indicates a certain stability in the tone of the *New York Times*, with no clear shift in sentiment across different administrations. Similarly, the year variable is not significant, suggesting that fluctuations in sentiment are more likely related to specific events rather than to a general temporal trend. Although the model explains only a small share of the variation (low adjusted R^2), the F-statistic confirms that the model is globally significant, indicating that the observed relationships are unlikely to be due to random chance.

Source of Variation	Df	Sum Sq (SS)	Mean Sq (MS)	F value	$Pr(>F)$	Decision
Year	1	0.1	0.078	0.1646	0.685	**Fail to Reject H_0**
President	4	12.3	3.066	6.4394	**3.581e-05**	**Reject H_0 *****
Word Count	1	39.7	39.677	83.3419	**< 2.2e-16**	**Reject H_0 *****
Residuals	12753	6071.3	0.476			

Figure 9: Test ANOVA results

Additional evidence is provided by the ANOVA test reported in Figure 9. The results confirm that word count significantly influences sentiment variation, reinforcing the finding that longer

abstracts tend to contain more negative language. The ANOVA test also shows that the presidential era significantly contributes to the overall variance in reporting**.** However, this effect does not translate into statistically significant coefficients in the regression model relative to the Bush baseline. Consistent with the regression results, the year variable is not significant, suggesting that variations in sentiment are mainly linked to specific events rather than long-term trends.

4.3 Topic modelling results

The model identified five main topics, each corresponding to a distinct dimension of migration coverage.

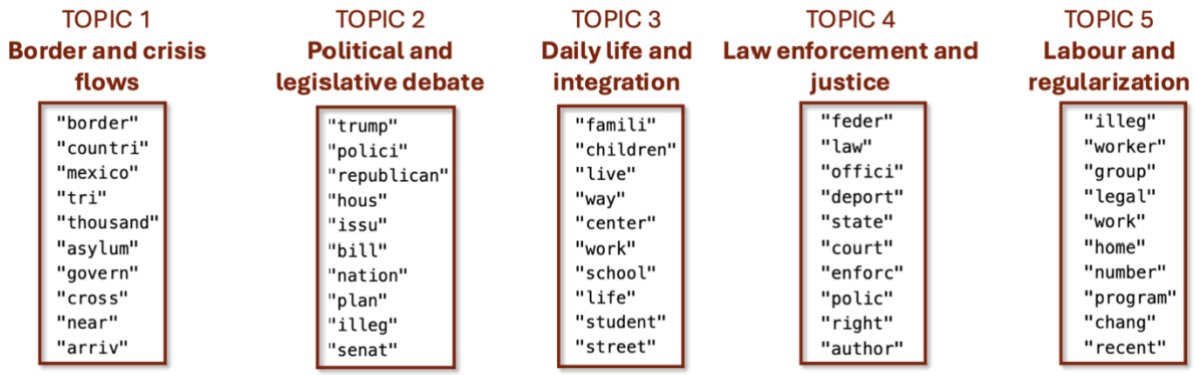


Figure 10: 5 main dominant topics.

The first topic, 'Border and Crisis Flows', focuses on the physical border and the logistical challenges associated with migration flows. Words associated with this topic refer to border crossings, geographical locations such as Mexico and emergency situations at the frontier. The second topic, 'Political and Legislative Debate', captures the institutional dimension of the migration discussion. It includes references to political actors, legislative bodies and policy debates taking place within the American political system. The third topic, 'Daily Life and Integration', represents a more social and human-centered perspective. It includes vocabulary related to families, schools, communities and the everyday experiences of migrants within American society. The fourth topic, **'**Law Enforcement and Justice', focuses on the legal and administrative mechanisms regulating migration, including courts, deportation procedures and federal enforcement agencies. Finally, the fifth topic, 'Labour and Regularization', highlights the economic dimension of migration, particularly the role of migrants in the labour market and discussions about legal status and workforce regularization.

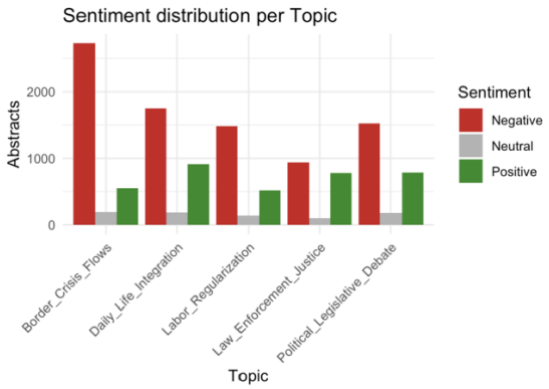


Figure 11: sentiment distribution per topic (AFINN).

Figure 11 illustrates the sentiment distribution across the five identified topics. Negative sentiment is the dominant frame across all categories, with ‘Border and Crisis Flows’ exhibiting the highest volume of negative abstracts. While ‘Daily Life and Integration’ shows a slightly higher proportion of positive reporting, it remains fundamentally skewed toward negativity, and neutral coverage is minimal across all themes.

Figure 12 details the specific words driving these sentiment scores. The term "illegal" acts as the primary negative driver in almost every topic, particularly in ‘Political Debate and Law Enforcement’. Furthermore, negativity is amplified by a vocabulary of conflict and emergency, including words like "arrest", "kill" and "war". Positive sentiment, driven by terms such as "support" and "hope" appears on a significantly smaller scale and is consistently outweighed by the negative lexicon.

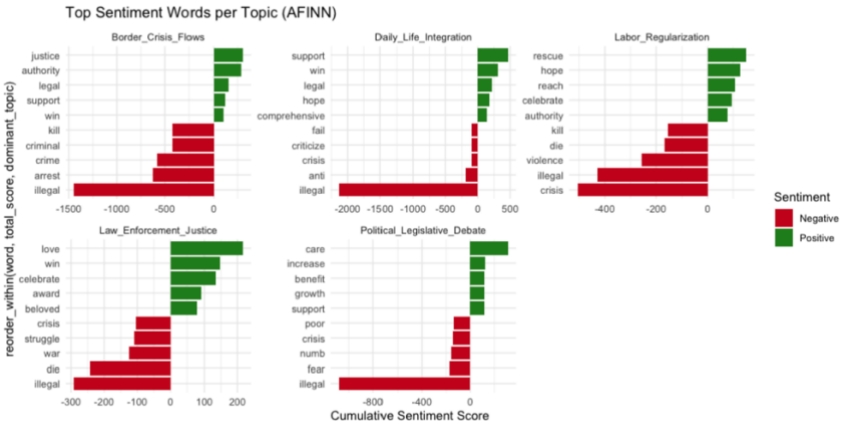


Figure 12: top sentiment words per topic (AFINN).

The longitudinal analysis (Figure 13) reveals that sentiment is not homogeneous across all migration themes. Specifically, "Border and Crisis Flows" remains consistently the most negative topic over the entire two-decade period, acting as the primary driver of the dataset's overall negativity. By contrast, when reporting shifts toward "Daily Life and Integration," the tone becomes slightly more balanced, though it never reaches true positivity. A particularly striking finding is the trajectory of the "Law Enforcement and Justice" theme. While it started as relatively neutral in 2005, it experienced a significant downward trajectory, plummeting into a high-intensity negative frame by 2025. This highlights a long-term narrative shift from treating migration as an administrative process to framing it through conflict and severity. Over time, most other themes, including political and labor discussions, have converged toward a similarly negative mean. Furthermore, this topic-level breakdown demonstrates that sudden fluctuations in overall sentiment are driven by border and justice events, rather than broad political discourse. Finally, the data confirms that presidential transitions do not cause substantial trend reversals, proving that this negative framing is a structural editorial feature rather than a partisan reaction.

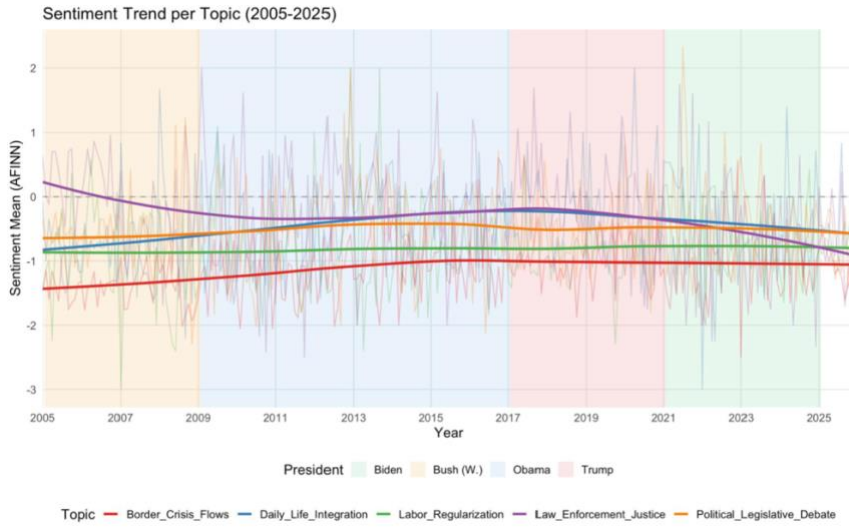


Figure 13: sentiment trend per topic (AFINN).

The multiple linear regression model demonstrates that incorporating thematic topics significantly increases the analysis's explanatory power, confirming that the specific subject matter is the primary driver of an article's tone. Specifically, the positive coefficients observed for themes like "Daily Life," "Law Enforcement," and "Political Debate" indicate that these topics are statistically less negative than the baseline topic, "Border and Crisis Flows". This reinforces the conclusion that border-related news constitutes the most critical and severe part of the overall narrative. Furthermore, word count remains a highly significant negative predictor, establishing that longer, more detailed articles consistently correlate with a harsher sentiment.

Predictor	Estimate (β)	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	-4.2366	7.8430	-0.540	0.589	
Year	0.0018	0.0039	0.486	0.627	
President (Obama)	-0.0050	0.0301	-0.169	0.866	
President (Trump)	0.0399	0.0505	0.791	0.429	
President (Biden)	-0.0738	0.0676	-1.091	0.275	
Word Count	**-0.0285**	**0.0033**	**-8.588**	**< 2e-16**	***
Topic: Daily Life	**0.2066**	**0.0174**	**11.872**	**< 2e-16**	***
Topic: Law Enforc.	**0.3460**	**0.0198**	**17.391**	**< 2e-16**	***
Topic: Political	**0.1309**	**0.0179**	**7.293**	**3.22e-13**	***

F-statistic: $p < 2.2*10^{(-16)}$ - **Adjusted R^2**: 0.0398

Figure 14: multiple linear regression model (with topic) summary.

Source of Variation	Df	Sum Sq (SS)	Mean Sq (MS)	F value	$Pr(>F)$	Decision (H_0)
Year	1	0.1	0.078	0.1700	0.6801	**Fail to Reject**
President	4	12.3	3.066	6.6526	**2.41e-05**	**Reject *****
Word Count	1	39.7	39.677	86.1014	**< 2.2e-16**	**Reject *****
Dominant Topic	4	196.4	49.108	106.5674	**< 2.2e-16**	**Reject *****
Residuals	12749	5874.9	0.461			

Figure 15: test ANOVA results.

While individual presidential administrations remain statistically non-significant in the regression, the subsequent ANOVA test provides additional nuance. It identifies the dominant topic as the single most influential factor in the overall model, immediately followed by article length. Additionally, the ANOVA reveals that the broader presidential era does introduce a statistically significant level of variance. This indicates that while the broader political context contributes to overall narrative fluctuations, no single president represents a radical departure from the newspaper's long-term negative editorial baseline.

5. Conclusions

This study shows that the migration narrative in The New York Times exhibits a persistent negative sentiment. This negativity constitutes more a structural feature of the reporting rather than a temporary trend. The analysis identifies the specific thematic focus, particularly border security or legal enforcement, as the primary predictor of sentiment. In addition, the results demonstrate a depth paradox: increased investigation does not provide a more balanced perspective, but instead correlates with more critical and severe reporting. Furthermore, the editorial baseline remains stable across the Bush, Obama, Trump and Biden time spans, indicating that the core sentiment lacks significant partisan shifts despite a polarized political landscape.

The computational framework applied in this research is reproducible and adaptable for analyzing articles across different decades and national contexts. However, some limitations can be found and improved. First, because the current study focuses only on The New York Times, future research should include diverse editors to allow for a comparative study of partisan media framing. Additionally, while abstracts provide a concentrated version of the narrative, examining full-text articles would yield a better understanding of the linguistic complexity and emotional depth of the coverage. Last but not least, comparing institutional newspaper narratives with social media discourse would help identify how migration sentiment differs between traditional journalism and public perception.

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